

A Randomized Rounding Approach for DAG Edge Deletion

Sina Kalantarzadeh, Nathan Klein, Victor Reis
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Presentation by: Mohammad Barekatin & Mohammad Reza Rahmani

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Preview

- Problem definition and hardness
- LP formulation
- Randomized rounding algorithm
- Feasibility and approximation analysis
- Improved bounds and special cases

Problem Definition

Problem (DAG Edge Deletion)

Given a directed acyclic graph (DAG) $G = (V, A)$, a weight function $w : A \rightarrow \mathbb{R}_+$, and an integer k , find a minimum-weight set $S \subseteq A$ such that, after deleting the edges in S , the graph contains no paths with k edges.

Note. DAG Edge Deletion is NP-hard for $k \geq 2$.

Example

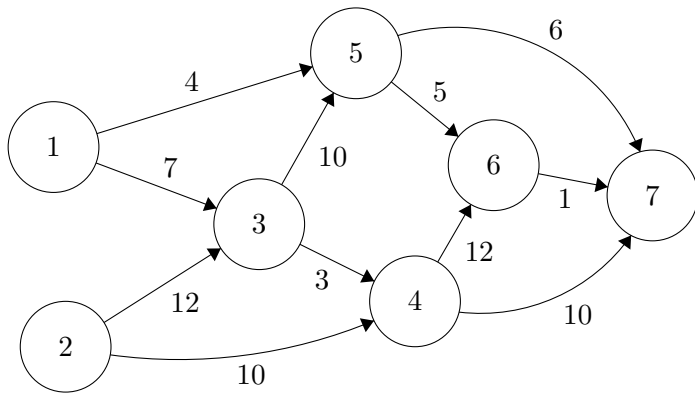


Figure: A directed acyclic graph with edge weights.

Example

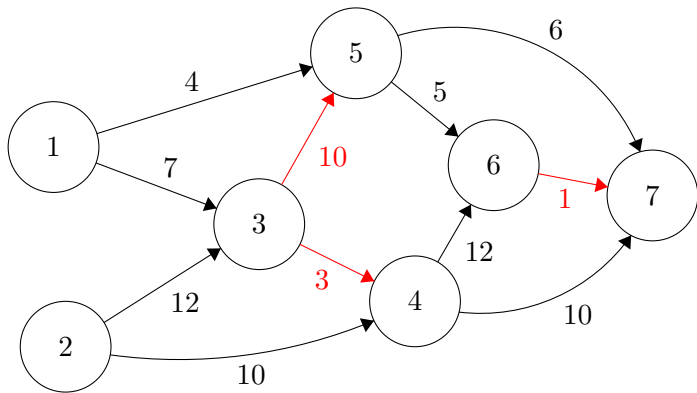


Figure: Solution to the example for $k = 3$. Optimal cost equals 14.

Example

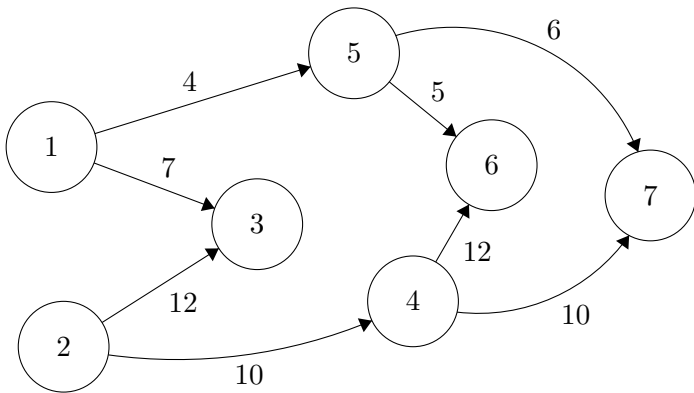


Figure: DAG after edge deletion has no path of length 3.

Related Work

Introduced by Kenkre, Pandit, Purohit, and Saket (2015):

- A simple deterministic rounding k -approximation algorithm
- UGC-hard to approximate within a factor better than $\lfloor \frac{k}{2} \rfloor$

Klein and Wexler (2016):

- A $\frac{2}{3}(k+1)$ -approximation algorithm using a combinatorial rounding approach
- An integrality gap of at least $\frac{k+1}{2}$ for natural LP

LP Formulation

The problem can be formulated as an integer program (IP) and relaxed to the following linear program (LP):

LP Relaxation

$$\begin{array}{ll} \min & \sum_{e \in A} w_e x_e \\ \text{s.t.} & \sum_{e \in P} x_e \geq 1, \quad \forall \text{ paths } P \text{ of length } k, \\ & x_e \geq 0, \quad \forall e \in A. \end{array} \quad (1)$$

Although the LP has exponentially many constraints, it can be solved in polynomial time using a separation oracle (via DP + DFS from each source).

Algorithm

Algorithm 1

1. Solve LP (1) to obtain values $\{x_e\}$
2. Assign each vertex u independently a label $l(u) \sim \text{Unif}(0, 1)$
3. For each edge $e = (u, v)$, set

$$\hat{x}_e = \begin{cases} 1 & \text{if } l(v) - l(u) \leq (k+1)x_e - 1, \\ 0 & \text{otherwise} \end{cases}$$

4. Return $S = \{e \in A \mid \hat{x}_e = 1\}$

Example

See the following example with $k = 2$. Suppose $w_e = 1$ for each edge. Clearly, setting $x_e = 0.5$ for all edges is an optimal feasible solution for LP (1).

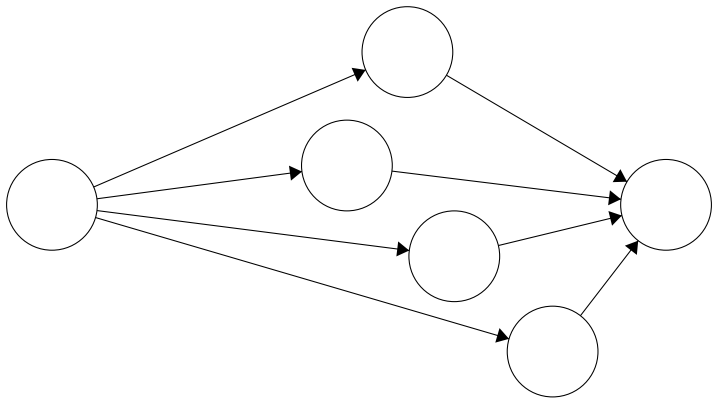


Figure: An example.

Example

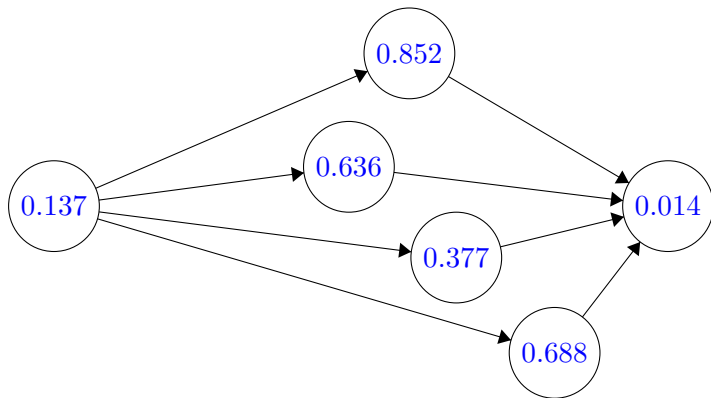


Figure: Assign random labels from a uniform distribution over $[0, 1]$.

Example

We remove an edge $e = (u, v)$, if $l(v) - l(u) \leq (2 + 1)(0.5) - 1 = 0.5$.

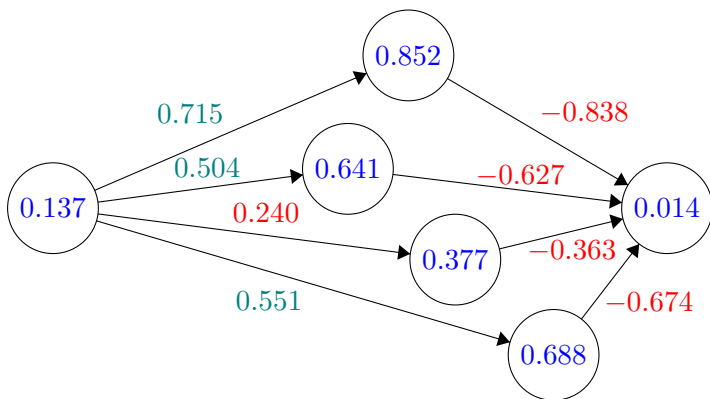


Figure: Calculate $l(v) - l(u)$.

Example

This leads to a feasible solution!

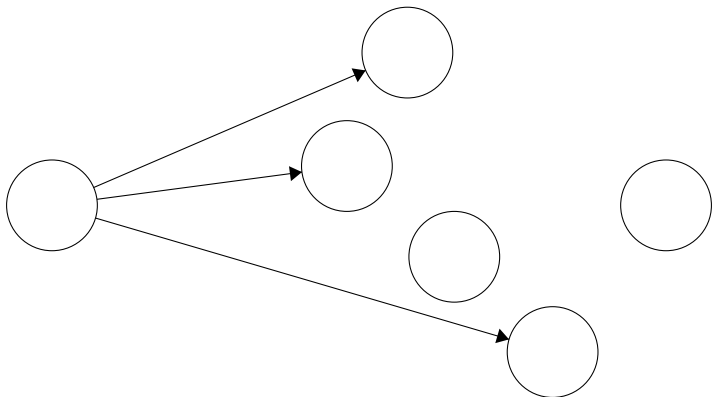


Figure: The resulting DAG has no path of length 2.

Feasibility

Lemma. For any feasible solution x to LP (1) and any labeling $l \in [0, 1]^V$, deleting all edges satisfying

$$l(v) - l(u) \leq (k + 1)x_e - 1$$

results in a feasible solution.

Proof (by contradiction).

Let P be a path of length k on vertices $v_1, \dots, v_{k+1}$. Assume that no edge on P is deleted. Then

$$l(v_{i+1}) - l(v_i) > (k + 1)x_{(v_i, v_{i+1})} - 1, \quad \text{for } 1 \leq i \leq k.$$

Feasibility

Proof continued.

Summing all inequalities,

$$l(v_{k+1}) - l(v_1) > \sum_{e \in P} [(k+1)x_e - 1].$$

Since $l(v_1) \geq 0$,

$$\begin{aligned} l(v_{k+1}) &> (k+1) \sum_{e \in P} x_e - k \\ &\geq (k+1) - k = 1, \end{aligned}$$

which contradicts $l(v_{k+1}) \in [0, 1]$. □

Analysis

Lemma. If labels are assigned independently and uniformly from $[0, 1]$, then for all $t \in [-1, 1]$,

$$\frac{\mathbb{P}[l(v) - l(u) \leq t]}{t + 1} \leq 2 - \sqrt{2}.$$

Proof.

If $l(u), l(v) \stackrel{i.i.d.}{\sim} \text{Unif}(0, 1)$, then the CDF of $l(v) - l(u)$ is

$$F(t) = \begin{cases} \frac{1}{2}t^2 + t + \frac{1}{2}, & -1 \leq t < 0, \\ -\frac{1}{2}t^2 + t + \frac{1}{2}, & 0 \leq t \leq 1. \end{cases}$$

Analysis

Proof continued.

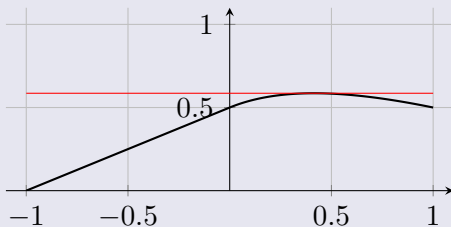


Figure: Plot of $y = \frac{F(t)}{t+1}$ (black) and $y = 2 - \sqrt{2}$ (red).

Hence, for all $t \in [-1, 1]$,

$$\frac{F(t)}{t+1} = \frac{\mathbb{P}[l(v) - l(u) \leq t]}{t+1} \leq 2 - \sqrt{2}.$$



Analysis

Theorem. The algorithm produces a solution with expected cost at most

$$(2 - \sqrt{2})(k + 1) \cdot w(x) \approx 0.585(k + 1) \cdot w(x),$$

yielding a $0.585(k + 1)$ -approximation.

Proof

Let $e = (u, v)$. We show

$$\mathbb{P}[l(v) - l(u) \leq (k + 1)x_e - 1] \leq (2 - \sqrt{2})(k + 1)x_e.$$

If $(k + 1)x_e - 1 \geq 1$, the inequality is trivial; otherwise, it follows from the previous lemma.

Analysis

Proof continued.

$$\begin{aligned}\mathbb{E}[w(S)] &= \sum_{e \in A} w_e \mathbb{P}[\hat{x}_e = 1] \\ &\leq (2 - \sqrt{2})(k + 1) \sum_{e \in A} w_e x_e \\ &= (2 - \sqrt{2})(k + 1) \cdot w(x).\end{aligned}$$



Thus, the approximation factor is $(2 - \sqrt{2})(k + 1)$. Can we do better?

Better Factor and Lower Bound

If the distribution of $l(v) - l(u)$ approaches $\text{Unif}(-1, 1)$, the approximation factor approaches $(k + 1)/2$.

- Using numerical methods, distribution $\mathcal{D}$ with PDF $p(x) = \frac{2}{3} + \frac{23}{3}(1 - 2x)^{22}$ is found, which yields a $0.549(k + 1)$ -approximation.
- It can be shown that this approach cannot achieve a factor better than $0.542(k + 1)$.

A better approximation factor may be possible for correlated labeling distributions.

Special Case: Bipartite Graphs

Theorem. Let $G = (V, A)$ be bipartite with bipartition (X, Y) and let x be a feasible solution to the LP. There is an algorithm that produces a solution with expected cost at most $0.5(k+1) \cdot w(x)$.

Proof

We use a correlated labeling distribution:

- Sample $y \sim \text{Unif}[0, 1]$
- With probability $\frac{1}{2}$:

$$l(v) = 0 \ \forall v \in X, \quad l(v) = y \ \forall v \in Y$$

- Otherwise:

$$l(v) = y \ \forall v \in X, \quad l(v) = 0 \ \forall v \in Y$$

Special Case: Bipartite Graphs

Proof continued.

For any edge $a = (u, v)$, the difference $l(v) - l(u)$ is uniformly distributed over $[-1, 1]$.

$$\begin{aligned}\mathbb{P}[a \text{ is cut}] &= \mathbb{P}[l(v) - l(u) \leq (k+1)x_a - 1] \\ &= \frac{(k+1)x_a - 1 + 1}{2} = 0.5(k+1)x_a\end{aligned}$$

Thus the expected total cost is:

$$\mathbb{E}[w(S)] = \sum_{a \in A} w_a \cdot 0.5(k+1)x_a = 0.5(k+1) \cdot w(x)$$



Any Questions?